



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO BHAKIYA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

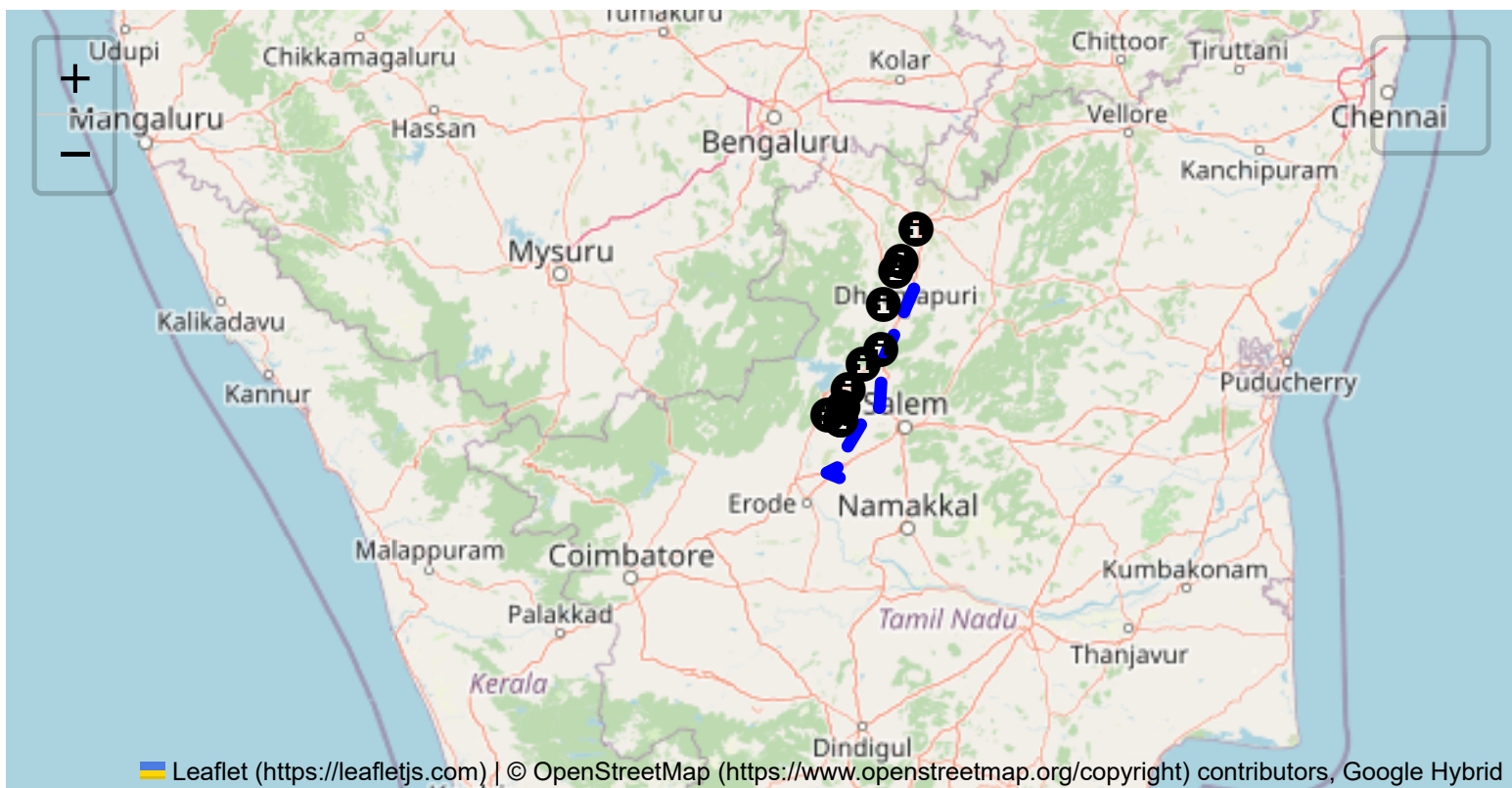
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

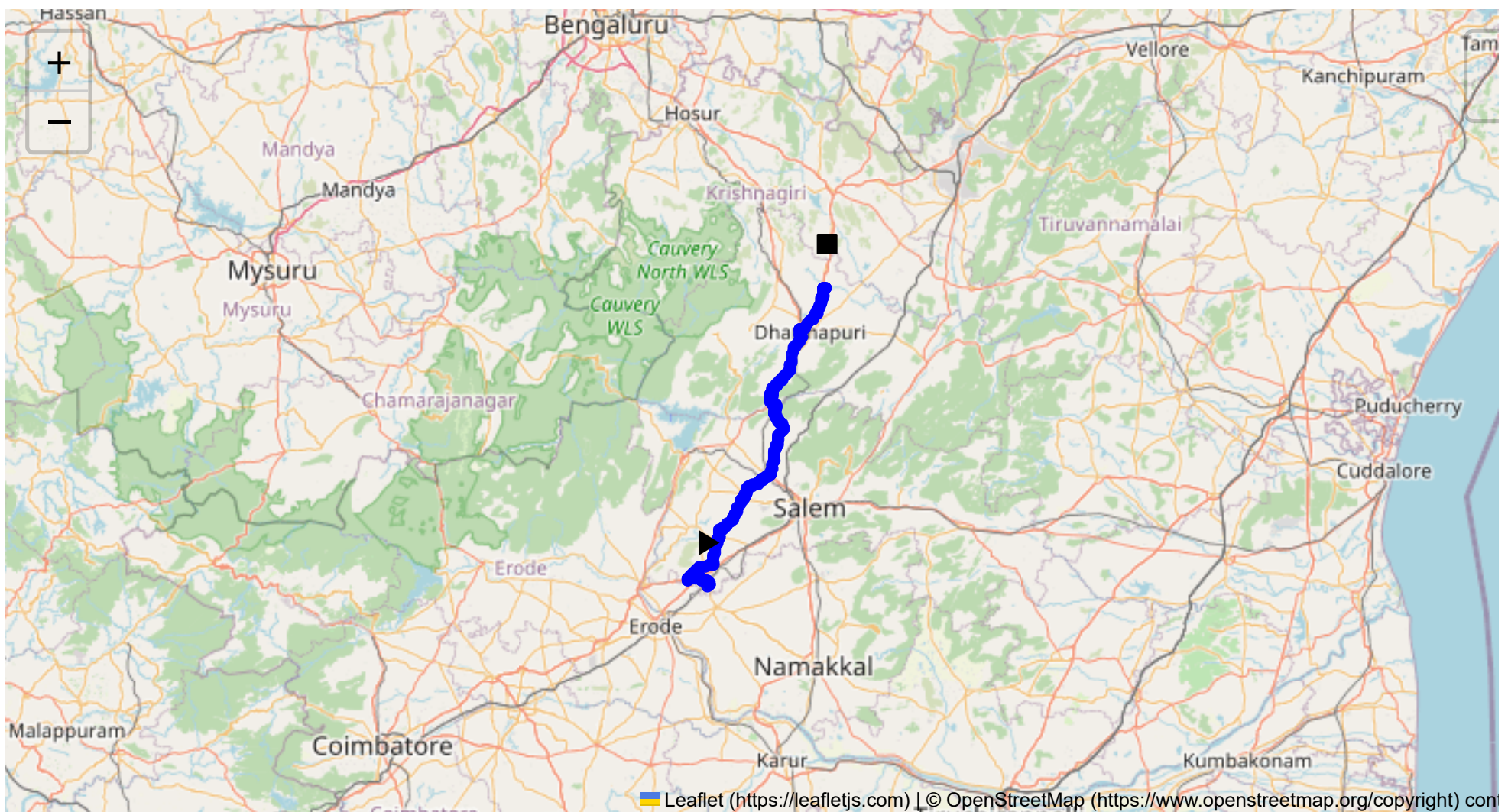
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 121.39 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours
Start: (11.4381, 77.8734)
End: (12.25404, 78.201939)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Adilam in Tamil Nadu covers approximately 121.39 kilometers and passes through key waypoints: Puthur, Bhavani Main Road, Salem - Kochi Highway, Omalur-Sankagiri Road, Desavilakku, Mettur Main Road, Thoppur, Adhiyaman Kottai, and Virupakshipuram. It includes a mix of state highways and national highways, which are generally well-connected with local road networks.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with hot summers, humid monsoons, and mild winters. Monsoon season, from June to September, can pose significant hazards due to heavy rain, leading to possible flooding and reduced visibility. The dry months are typically devoid of major weather disturbances, but heat can impact vehicle performance.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, traffic is heavier during morning (7-10 AM) and evening (5-8 PM) hours on weekdays, especially near urban areas and highway junctions.
- **Congestion-Prone Areas:** Significant congestion is often observed at:
 - Salem - Kochi Highway junctions
 - Omalur Branch areas
 - Entry and exit points in Thoppur and Dharmapuri

4. Assessment of Road Quality and Infrastructure

- Road quality is generally good on highways, but some segments, particularly rural byways near Padaiveedu and Erumaipatti, may have potholes and uneven surfaces.
- Infrastructure such as signage, lighting, and available rest areas are sufficient but could be improved in less developed zones.

5. Suggestions for Alternative Routes for Emergencies

Alternative routing could include:

- Using NH-44 as a bypass for heavy congestion around Salem.
- Route diversions toward interior roads near Erumaipatti in case of highway blocks or accidents.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Vehicle compliance with the National Permit System is mandatory.
- Restricted movements during peak hours within city limits apply.
- Specific regulations on hazardous material signage and documentation are enforced.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Past incidents primarily include road mishaps due to human error or vehicle malfunction, with fewer reports of severe hazardous material breaches. Historical accident data suggest an increased risk during monsoon due to slippery roads.

8. Environmental Considerations and Sensitive Areas

- Nearby farmlands and water bodies could be adversely impacted by spills.
- Certain sections through forested areas near Thoppur are sensitive due to wildlife presence, requiring careful navigation and adherence to speed limits.

9. Analysis of Communication Coverage

While major towns and highways have robust cell coverage, rural segments may experience dead zones, particularly in areas between Erumaipatti and Desavilakku.

10. Estimated Emergency Response Times

- Urban areas like Salem: 15-30 minutes
- Rural or less accessible areas: up to 45-60 minutes due to logistics and road conditions

12. Overall Summary of Risk Assessment

The route is generally safe for trucking hazardous materials when adhering to safety protocols and regulations. Weather conditions during monsoon pose the greatest risks alongside traffic congestion. Clear communication systems and emergency protocols need to be established and followed, particularly in remote areas. Environmental considerations are vital, especially in sensitive zones. Advanced planning for alternative routing and ensuring vehicle compliance with regional laws are integral parts of risk management.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45543, 77.81740	30 KM/Hr
8	Turn	High	11.45631, 77.81668	15 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.58254, 77.90656	30 KM/Hr
13	Turn	Medium	11.63867, 77.94841	30 KM/Hr
14	Turn	Medium	11.64217, 77.95200	30 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	High	11.67858, 77.96834	15 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72559, 78.01758	30 KM/Hr
21	Turn	Medium	11.73787, 78.03528	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74257, 78.04708	30 KM/Hr
24	Turn	Medium	11.86805, 78.08353	30 KM/Hr
25	Turn	Medium	11.87913, 78.08206	30 KM/Hr
26	Turn	Medium	11.92566, 78.06503	30 KM/Hr
27	Turn	Medium	11.93694, 78.05425	30 KM/Hr
28	Turn	Medium	11.95531, 78.04698	30 KM/Hr
29	Turn	Medium	11.95854, 78.04990	30 KM/Hr
30	Turn	Medium	11.96068, 78.04960	30 KM/Hr
31	Turn	Medium	11.96370, 78.05142	30 KM/Hr
32	Turn	Medium	11.96433, 78.05113	30 KM/Hr
33	Turn	High	11.96542, 78.04938	15 KM/Hr
34	Turn	High	11.96650, 78.04927	15 KM/Hr
35	Turn	Medium	11.96804, 78.05080	30 KM/Hr
36	Turn	Medium	12.12017, 78.13283	30 KM/Hr
37	Turn	High	12.14089, 78.12817	15 KM/Hr
38	Turn	High	12.14056, 78.13957	15 KM/Hr
39	Turn	High	12.25431, 78.19669	15 KM/Hr
40	Turn	Medium	12.25448, 78.20005	30 KM/Hr
41	Turn	High	12.25404, 78.20187	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58254, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72559, 78.01758



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708

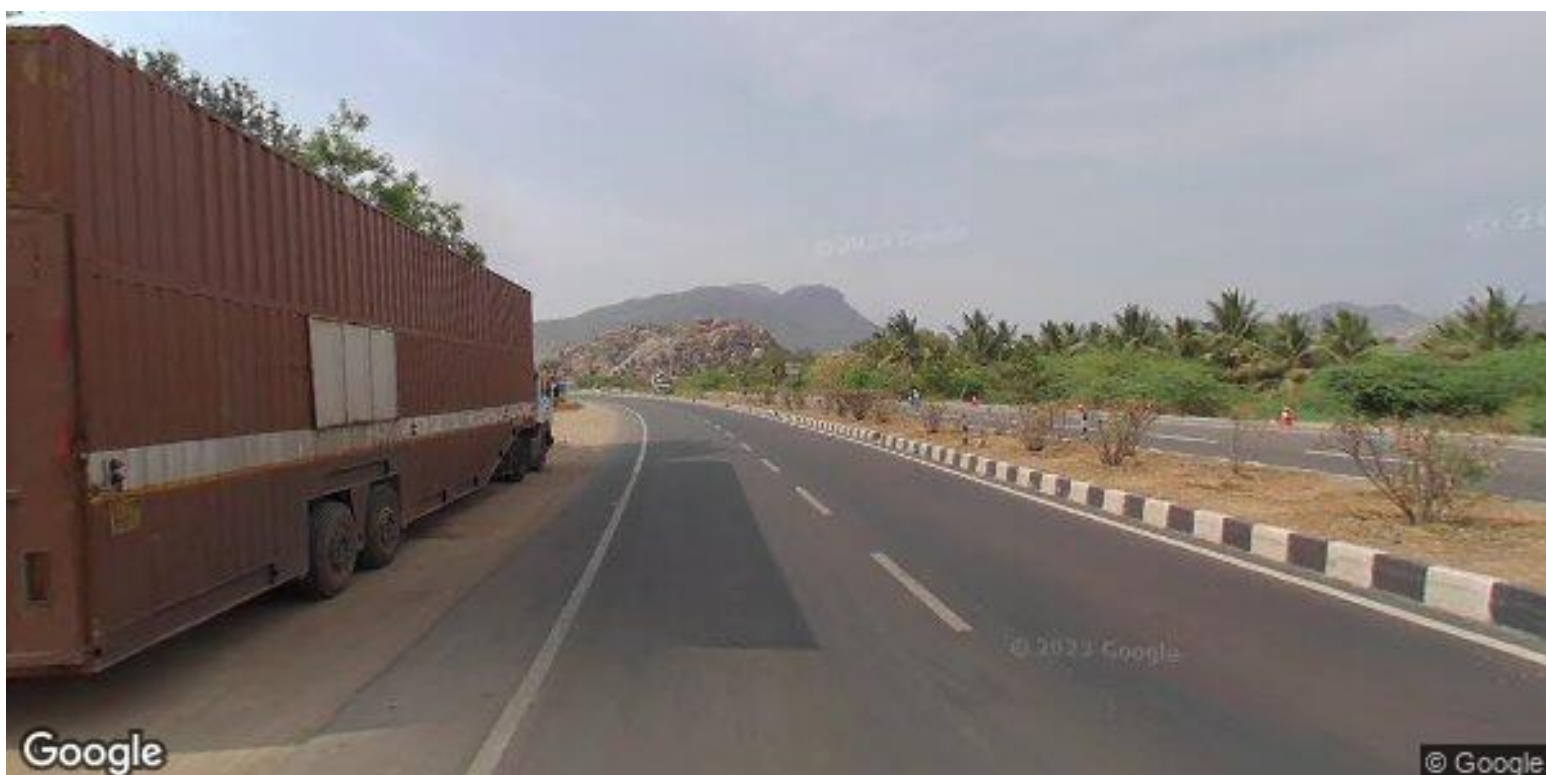


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92566, 78.06503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95531, 78.04698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96068, 78.04960



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96542, 78.04938



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12017, 78.13283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14089, 78.12817



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14056, 78.13957



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.25431, 78.19669



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25448, 78.20005



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.25404, 78.20187